



SMARTPHONE SPECIFICATION TREND ANALYSIS USING PYTHON

SUBMITTER BY :- SEJAL

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COMPANY PROFILE



Certification: ISO 9001:2015 Certified

Specializations:

- Web Development
- Mobile Application Development
- Custom Software Development
- UI/UX Design
- Hosting Services
- Digital Marketing

Services:

- Advanced IT solutions supporting the entire business
- Consulting to system development

Headquarters: Jalandhar

Branch Office: Hoshiarpur

COMPANY PROFILE



Company Name: O7 Solutions IT

Company

Founded: 2017

Certification: ISO 9001:2015 Certified

Specializations:

- Mobile Development
- Web Development
- Software Development
- Digital Marketing
- IT Integration

Experience: Over 7 years

Drawbacks of Existing System

Unstructured Data :- Hard to understand without cleaning.

No Visual Insight :- No clear comparison Between brand and features.

Manual & Time-Consuming:- Difficult to find Best value Phone without Analysis.

No clear Price Analysis:- Smartphone price could not be compared by features or brands.

INTRODUCTION

- Analyzes how smartphone features affect pricing
- Key attributes: RAM, Storage, Battery, Camera, Display
- Used Python libraries: Pandas, NumPy, Matplotlib
- Helps consumers and businesses understand price-driving factors
- Identifies trends and correlations in pricing
- Helps consumers choose better-value smartphones

Objective / Solution

Analyze
smartphone
data using
Python and data
science libraries.

Identify which
features (RAM,
Camera,
Battery, etc.)
impact price.

Visualize
relationships
between specs
and pricing
using charts.

Compare pricing
trends across
different mobile
brands.

MODULE

Data Collection :-

Gather smartphone specs and pricing data from reliable sources (e.g., Kaggle).

Data Preprocessing :-

Clean, format, and structure the raw data for analysis.

Exploratory Data Analysis (EDA) :- Use graphs to find trends (e.g., Price vs. RAM, Storage, Brand).

Visualization :- Create charts and plots using Python libraries (Matplotlib, Seaborn).

Result Interpretation :-

Analyze findings to understand what features impact price the most.

Conclusion & Insights

Help users and businesses make informed decisions based on results.

Advantages of Project

Price Comparison Made Easy :- Users can compare prices across brands and specifications quickly.

Informed Buying Decisions :- Consumers choose the best smartphone based on RAM, storage, battery, etc.

Market Trend Analysis :- Identifies trends by brand, features, or region.

Fast Insights Through Graphs :- Visuals make understanding complex data quick and simple.

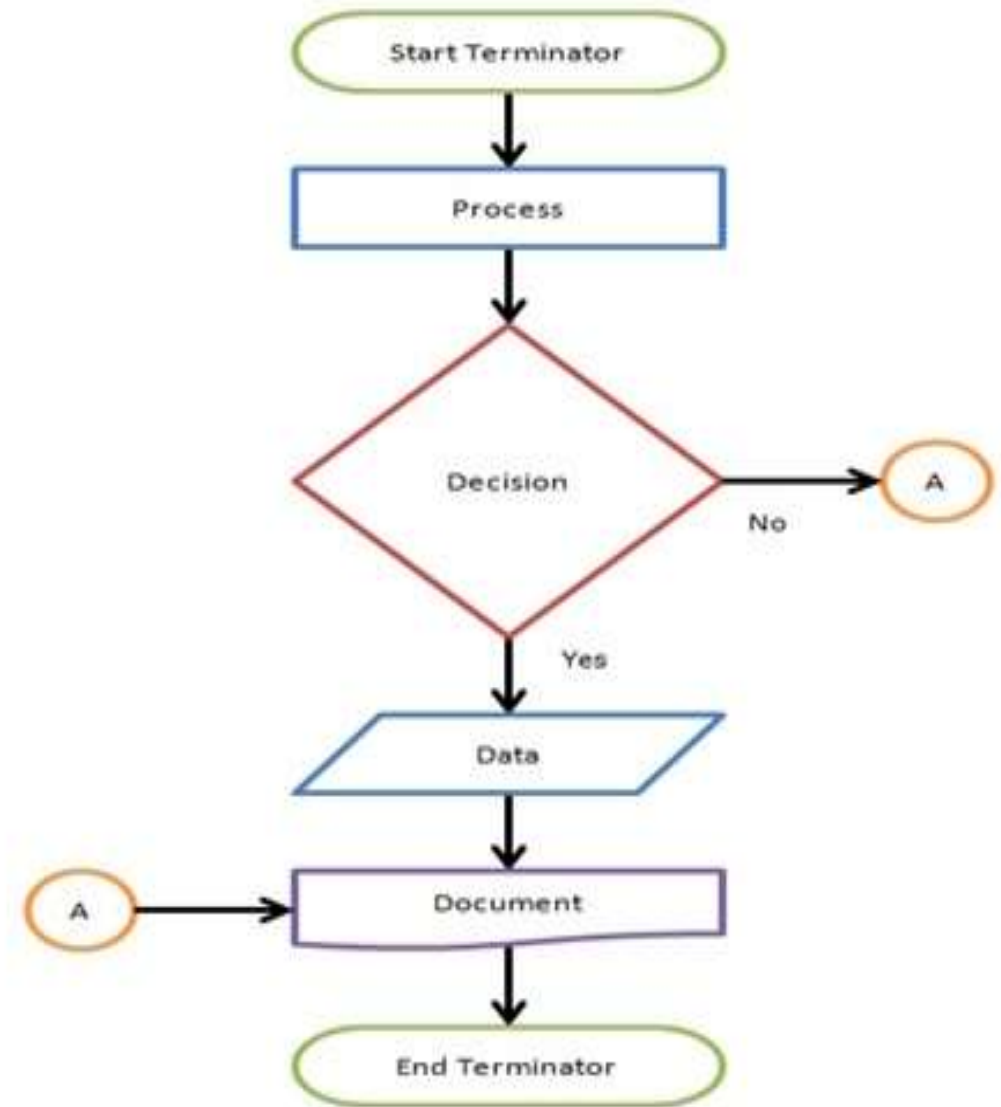
Learning Outcome for Analysts :- Gain hands-on experience with Python, Pandas, Matplotlib, and real-world datasets.

Tools & Technologies

Layer	Technology	Purpose
Data Analysis	Python	Core language for analysis, scripting, and automation
Libraries	Pandas	Data manipulation, filtering, cleaning
	NumPy	Numerical operations, array handling
	Matplotlib	Visualization using bar, line, scatter plots
	Seaborn	Advanced statistical graphs and plots
Environment	Jupyter Notebook	Interactive coding and analysis
Dataset Source	Kaggle	Public dataset for mobile specs and prices

FLOWCHART

- A flowchart is a picture of boxes that indicates the process flow sequentially.
- It is a pictorial way of representing steps that are preferred by most beginner-level programmers to understand algorithms of computer science



Dataset Overview

❖ Source of Dataset

Platform :- Kaggle

Dataset Name :- Mobile Phones Specifications and Prices

❖ Covers

Global smartphone specifications

Launch prices across countries

Technical hardware details

Key features like camera, display, battery, storage

❖ Purpose of Dataset

Price comparison & market trend analysis

Data analysis and machine learning use

Data Flow Diagram (DFD)

Here is the meaning of each DFD (Data Flow Diagram) symbol explained In points:

External Entity – Outside user or system (e.g., User giving input)

Process – Action performed by the system (e.g., Mobile Price Analysis)

Data Store – Saves data like databases (e.g., Smartphone Specs)

Data Flow – Shows how data moves between parts (e.g., User Process)

DFD – Data Flow Diagram

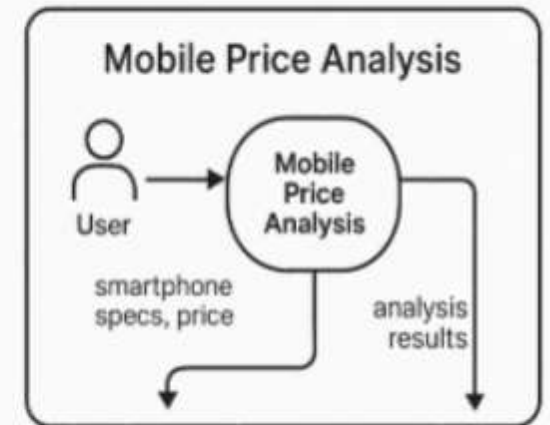
A Data Flow Diagram (DFD) is a visual representation of how data moves through a system.

👤 External Entity

○ Process

🗄️ Data Store

→ Data Flow



DATA CLEANING

Handled Null Values:

Missing values were filled using mean/median or removed if not useful.

Data Type Conversion:

Converted string-based numbers (e.g., "6GB") into proper numeric types.

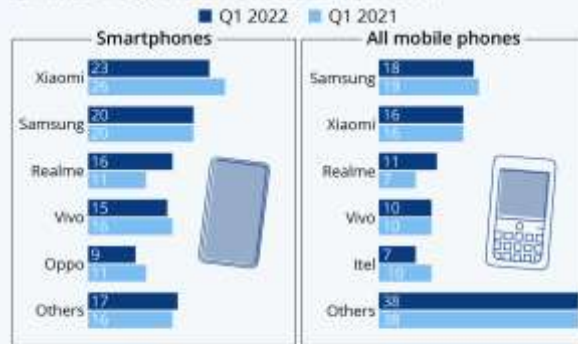
Removed Duplicates:

Deleted duplicate rows to avoid repetition in analysis.

Reindexing & Renaming: Reindexed rows and renamed columns for better readability.

China Dominates Indian Mobile Phone Market

Mobile phone shipment market shares in India in Q1 of 2021 and 2022 (in percent)



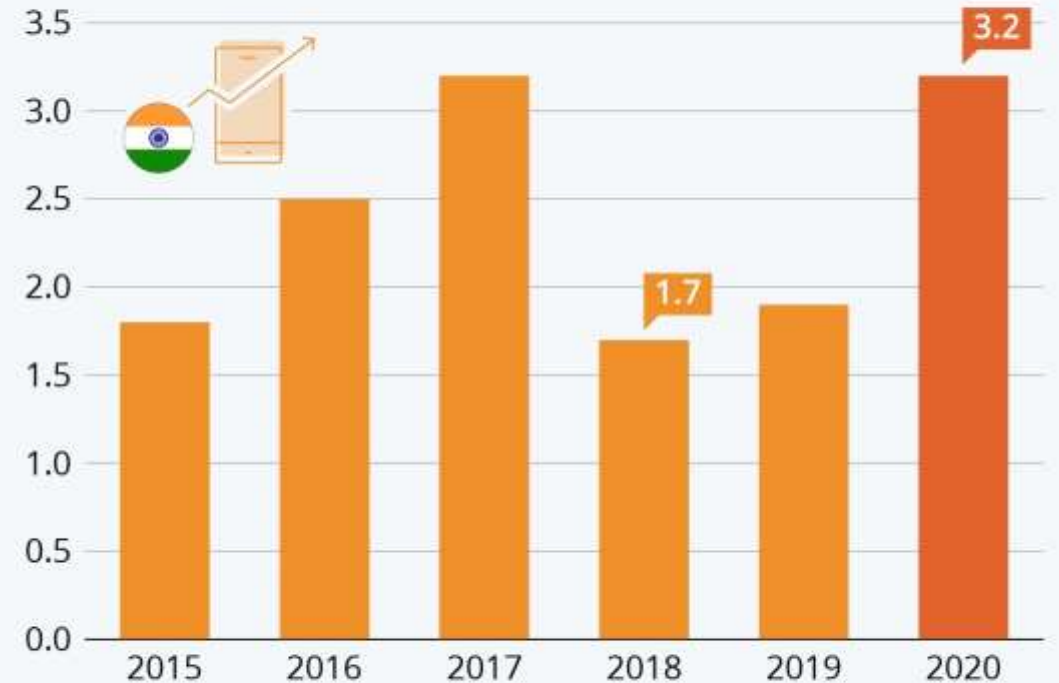
Source: Counterpoint Research



statista

Demand for iPhones up in India

Number of iPhones shipped in India by Apple (in millions)



Source: Counterpoint Research



statista



Future Scope and Enhancements

Add More Features :- Include parameters like 5G support, refresh rate, OS version, waterproof ratings

Price Prediction Model :- Use Machine Learning (e.g., Linear Regression, Decision Tree) to predict mobile prices based on specs

Dynamic Dashboard :- Create a real-time interactive dashboard using Power BI or Tableau

Platform Development :- Convert this analysis into a user platform with vendor and admin roles

Global Comparison :- Expand dataset with international pricing and currency normalization

Conclusion

- Features like RAM, Storage, Battery Capacity, Camera Quality, and Display Size significantly impact the launch price of smartphones.
- Brand reputation plays a crucial role—premium brands generally have higher price ranges despite similar hardware.
- Through data preprocessing, we ensured the dataset was clean, structured, and ready for meaningful analysis.
- Visualizations (like bar charts, correlation heatmaps, and distribution plots) made it easy to interpret complex patterns.
- Identified budget vs premium segment trends, which can help both users and sellers understand the market better.



THANK YOU